

IDS 570: Text as Data
Data Exploration Assignment
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Political Economy Corpus - Early Modern English Trade Texts

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1. Introduction and Corpus Description

This report presents a structured exploration of a 23-document corpus of early modern English political economy texts, spanning approximately 1580–1776. The corpus includes works from several overlapping intellectual traditions: monetary debate texts associated with Gerard de Malynes (c. 1601–1623); Misselden’s Circle of Commerce (1623); colonial and plantation pamphlets debating poor relief, usury, and trade regulation from the mid-to-late seventeenth century; East India Company tracts; and Adam Smith’s *Wealth of Nations* (1776) as a temporal outlier and baseline for comparison.

Three analytical approaches were applied in sequence.

1. TF-IDF weighting identified lexically distinctive terms for each document.
2. Pearson correlation coefficients computed on a trimmed document-feature matrix quantified pairwise similarity across the corpus.
3. five parser-free surface measures of register complexity - mean sentence length (MSL), mean word length (MWL), moving-average type-token ratio (MATTR), punctuation density (PD), and long-word ratio (LWR) - were computed across all 23 documents.

2. Normalization Decisions

The corpus required several normalization steps before analysis. One was mandatory; three were optional but principled.

The mandatory step was long-S replacement: EEBO transcriptions render the early modern long S character (ſ) as a distinct Unicode codepoint. Left unresolved, tokens like “buſineſ” and “business” would be counted as distinct types. All instances were replaced with standard ‘s’ prior to tokenization.

The optional steps were:

1. removal of non-Latin alphabet placeholder tags (<in non-Latin alphabet>), representing Greek and Hebrew citations rendered as markup fragments in the EEBO XML pipeline.
2. removal of typographic artefact characters (• ▪ ◇) used as marginal indicators in some transcriptions; and
3. addition of an early modern English stopword list supplementing the standard English stopword list, covering archaic function words such as ‘hath’, ‘doth’, ‘vnto’, ‘haue’, and

similar forms that carry no lexical content but would otherwise dominate frequency counts.

Two decisions were deliberately not made: u/v alternation (e.g. ‘vpon’/‘upon’) and variant spellings of substantive terms (e.g. ‘mony’/‘money’, ‘moneys’/‘monies’) were preserved. Collapsing these would obscure potentially meaningful distinctions: variant spelling patterns may reflect authorial habit, scribal convention, temporal change, or genre differences, all of which are relevant to the analytical questions pursued here.

A final post-tokenization cleaning step removed abbreviated reference fragments that survived punctuation removal (‘ll’, ‘ss’ from accounting tables; ‘arg’, ‘answ’ from legal marginal abbreviations) and all tokens under four characters, which in this corpus are predominantly typographic rather than lexical.

3. Approach 1: TF-IDF- Lexical Distinctiveness

3.1 Method

TF-IDF (term frequency-inverse document frequency) measures how distinctive a term is to a given document relative to the corpus. A term appearing frequently in one document but rarely across others receives a high TF-IDF weight; common terms shared across many documents receive low weights regardless of local frequency. The top 12 terms per document were extracted from a TF-IDF-weighted document-feature matrix built using the quanteda package in R, after applying stopword removal and post-tokenization artefact cleaning.

3.2 Results

Document	Top 12 TF-IDF Terms
A06785	moneys, misselden, exchange, circle, realm, doller, undervaluation, royals, stivers, overbalancing, commodities, centre
A06786	hundreth, ounces, weight, carrats, moneys, factor, assurors, grains, antuerp, silver, unto, ducats
A06788	sols, silver, livers, realm, dearth, commodities, gold, bodine, price, moneys, velvet, shillings
A06789	moneys, monies, exchange, realm, commodities, shillings, doller, enhancing, price, stivers, pence, albeit
A06790	monster, dragon, moon, unto, tail, sweet, causeth, leaguors, smell, feeding, visions, hell
A06791	realm, moneys, commodities, exchange, fineness, price, shillings, silver, exchangers, albeit, hundredth, bank
A07594 (Circle of Commerce)	malynes, kingdome, mony, ballance, forraine, owne, lesse, forme, himselfe, gaine, custome, sayd

Document	Top 12 TF-IDF Terms
A07886	wares, raw-silkes, realm, indigo, indies, aleppo, rawsilke, india, yearly, east, kingdom's, sterling
A32827	plantations, abatement, new-england, poor, usury, interest, english, newfoundland, laws, planting, barbadoes, judiciary
A32828	support, mortgage, disposal, purely, promotion, impediments, commerce, advantages, creditors, produce, ministering, instrumental
A32829	packs, stuffs, cloth, wool, broad-cloths, merchant-adventurers, manufactors, foreigners, thousand, growers, african, price
A32830	taffities, muslins, product, india, prime, persuade, improvement, expended, mony, scotch, undeniable, puts
A32833	plantations, abatement, new-england, poor, usury, interest, english, newfoundland, laws, planting, barbadoes, planters
A32836	poor, parish, fathers, propose, hospitals, pious, assembly, question, quest, elected, election, defect
A32837	usurer, usurers, barbadoss, objection, planters, richest, plantations, use-money, qualifications, lender's, interest, territory
A32838	surrat, phirmaund, mogul, bombay, mogul's, guns, india, english, mary, committees, cargoes, kempthorne
A32839	tuns, east-india, seamen, protestant, india, charter, bengall, company, regulated, suratt, silk, legitimate
A50763	poor, parish, fathers, goals, propose, hospitals, assembly, design, communication, elected, election, dorchester
A51598	wares, treasure, genova, foreign, moneys, realm, exchange, gain, divers, feats, country's, wonders
A69858	india, surrat, east-india, bantam, tuns, company, charter, bombay, naval, deck, legitimate, bengall
A93819	lion-key, stairs, edmond, wiseman, defendant, wharfige, london-bridge, surveyors, thames, enrolled, wharves, wharf
B14801 (Free Trade)	kingdome, common-wealth, forreine, gouvernement, maiesties, subiects, vnder, christendome, publique, owne, selfe, euery
Wealth	labor, however, capital, revenue, expense, system, productive, produce, different, century, rent, frequently

Table 1. Top 12 TF-IDF terms by document, ranked by descending weight.

Figure 1 displays these results visually. Each panel uses an independent x-axis scale, so that documents with extreme TF-IDF values (notably A06786 and wealth) do not visually suppress all other documents.



Figure 1. Top 12 TF-IDF terms per document. Each panel has a free x-axis scale.

3.3 Interpretation

The Malynes monetary texts (A06785, A06786, A06788, A06789, A06791) share a dense cluster of high-weight terms: ‘moneys’, ‘realm’, ‘exchange’, ‘commodities’, ‘price’, ‘shillings’. This lexical coherence reflects their shared analytical framework, engaging with bullion flows, exchange rates, and commodity pricing. Specific currency units (‘ducats’, ‘stivers’, ‘doller’, ‘sols’) and place names (‘antuerp’, ‘genova’) distinguish individual texts within this cluster.

A06790, Malynes’s *Canker of England’s Commonwealth* (1601), is the most striking outlier among the Malynes texts. Its top terms - ‘monster’, ‘dragon’, ‘moon’, ‘tail’, ‘smell’, ‘feeding’, ‘visions’, ‘hell’ - reveal an allegorical mode entirely different from the analytical register of the other monetary texts. The *Canker* is an extended moral allegory on usury cast in beast-fable form, and its lexical signature reflects genre rather than technical economic content. The presence of ‘unto’ in its top terms is a residual artefact of the high frequency of this archaic function word in this text; it was not fully suppressed by the stopword list, suggesting it should be added to future iterations.

Misselden’s *Circle of Commerce* (A07594) presents another interpretively rich result. Its highest-weighted term is ‘malynes’ (TF-IDF = 140), confirming that the text is structured as a point-by-point refutation of Malynes rather than an independent treatise. The appearance of ‘forraigne’, ‘ballance’, and ‘kingdome’ alongside this shows how polemical engagement compels Misselden to adopt and contest Malynes’s own technical vocabulary.

The colonial and plantation pamphlets (A32827, A32828, A32829, A32830, A32833, A32836, A32837) cluster around ‘plantations’, ‘poor’, ‘usury’, ‘interest’, ‘parish’, and ‘abatement’, reflecting overlapping debates around poor relief, colonial settlement, and the lawfulness of lending. The East India Company texts (A07886, A32838, A32839, A69858) are unified by trade geography: ‘surrat’, ‘india’, ‘bengall’, ‘bantam’, ‘east-india’. The distinctiveness of ‘phirmaund’ and ‘mogul’ in A32838 identifies that text’s focus on diplomatic relations with Mughal authority specifically.

The wealth document (Adam Smith’s *Wealth of Nations*) stands lexically apart from all early modern texts. Its top terms - ‘labor’, ‘capital’, ‘revenue’, ‘system’, ‘productive’, ‘century’ - reflect both its eighteenth-century date and the conceptual vocabulary of classical political economy, which was still being formed during the period of the other corpus documents. The high absolute TF-IDF values for these terms (labor = 1071; capital = 383) partly reflect document length: at 380,080 tokens, wealth is over 130 times larger than the smallest corpus texts.

4. Approach 2: Pearson Correlation - Document Similarity

4.1 Method

Pairwise Pearson correlation coefficients were computed between all 23 documents on a trimmed document-feature matrix (minimum term frequency = 5 across the corpus). Trimming removes hapax legomena and near hapax tokens whose sparsity would otherwise inflate correlations between documents sharing rare terms by chance. The resulting 23×23 matrix was visualized as a heatmap; the five most and least similar pairs were identified.

4.2 Results

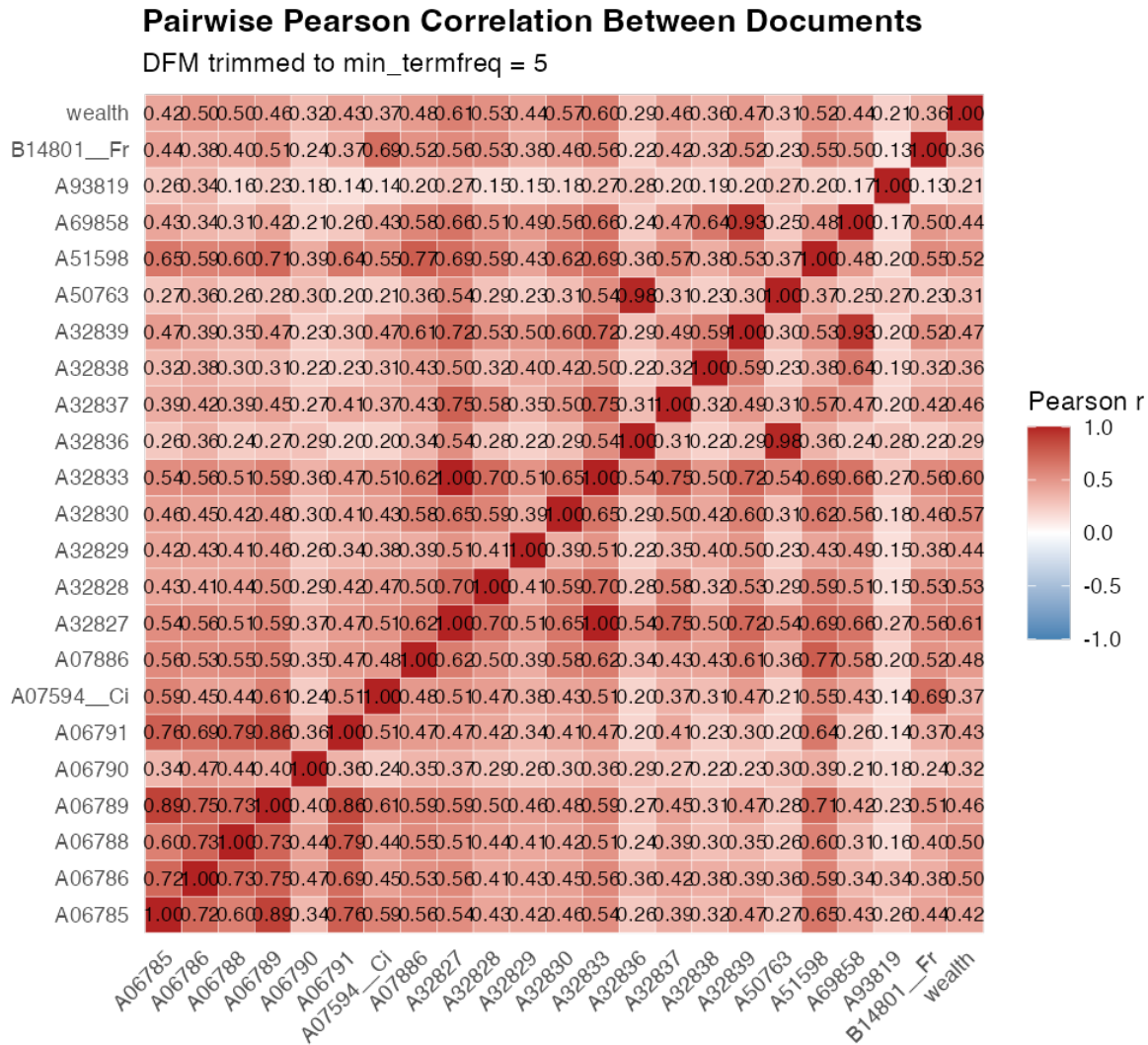


Figure 2. Pairwise Pearson correlation matrix. Darker red = higher similarity.

Five most similar pairs:

1. A32827 – A32833: $r = 1.000$
2. A32836 – A50763: $r = 0.983$
3. A32839 – A69858: $r = 0.929$
4. A06785 – A06789: $r = 0.886$
5. A06789 – A06791: $r = 0.857$

Five least similar pairs:

1. A93819 – B14801 (Free Trade): $r = 0.126$
2. A07594 (Circle of Commerce) – A93819: $r = 0.136$
3. A06791 – A93819: $r = 0.143$
4. A32828 – A93819: $r = 0.148$
5. A32829 – A93819: $r = 0.151$

4.3 Interpretation

The most striking result is $r = 1.000$ between A32827 and A32833. Inspection of the TF-IDF terms confirms these are near-identical texts: both have identical top terms (plantations, abatement, new-England, poor, usury, interest, English, newfoundland) with virtually identical weights and share nearly the same token count (49,656 vs 49,725).

They are almost certainly two editions or printings of the same petition on trade and plantations. Similarly, A32836 and A50763 ($r = 0.983$) share top terms of ‘poor’, ‘parish’, ‘fathers’, ‘hospitals’, ‘elected’, and ‘election’, identifying them as near-duplicate versions of the same poor-relief proposal.

Setting aside these near duplicates, the highest genuine pairwise similarities are among the Malynes monetary texts: A06785/A06789 at $r = 0.886$, and A06789/A06791 at $r = 0.857$. This confirms the TF-IDF finding that these texts share a dense common technical vocabulary. The broader Malynes cluster (A06785, A06786, A06788, A06789, A06791) is clearly visible as a dark block in the upper-left of the heatmap.

A93819 is the most isolated document in the corpus, appearing in all five least-similar pairs and recording correlations below $r = 0.28$ with all but one other document. Its TF-IDF terms (‘lion-key’, ‘wharfage’, ‘stairs’, ‘surveyors’, ‘thames’) identify it as a legal deposition concerning wharfage rights on the Thames - a genre entirely different from all others in the corpus. Its low correlations are not evidence of unusual content within the tradition of commercial writing; rather, the text is simply not a commercial writing text in the relevant sense.

A notable cross-author finding is the moderate correlation between Misselden’s Circle of Commerce (A07594) and the Malynes monetary treatises: $r = 0.592$ with A06785 and $r = 0.609$ with A06789. These are higher than Misselden’s correlation with the colonial pamphlets or East India texts, consistent with the TF-IDF finding that polemical engagement produces lexical convergence: Misselden must adopt Malynes’s technical vocabulary to contest it.

The East India Company texts form a tight sub-cluster: A32839/A69858 at $r = 0.929$ and A32838/A32839 at $r = 0.593$. The wealth document (Adam Smith) occupies a middling position across the matrix ($r \approx 0.4$ – 0.6 with most texts), consistent with its status as a much later retrospective synthesis that draws on the vocabulary of its predecessors while deploying a newer conceptual framework.

5. Approach 3: Surface Register Complexity

5.1 Method and Rationale

The third approach measures register complexity using five surface-level features computed directly from tokenized text, requiring no syntactic parser. This choice is principled: the corpus consists almost entirely of early modern English texts (c. 1580–1700), and dependency parsers trained on modern English introduce unknown error rates when applied to this period’s grammar, orthography, and punctuation conventions. Surface measures derived directly from tokenization are therefore more reliable for cross-document comparison in this corpus.

Five measures were computed across all 23 documents from the normalized raw texts. Mean Sentence Length (MSL) counts word tokens per punctuation-delimited sentence. Mean Word Length (MWL) averages character counts per token, serving as a proxy for Latinate versus Germanic vocabulary balance. Moving-Average Type-Token Ratio (MATTR), computed over a

sliding window of 100 tokens, measures lexical diversity independently of document length, unlike simple TTR which degrades for long texts. Punctuation Density (PD) counts punctuation marks per 100 tokens, reflecting syntactic segmentation habits. Long-Word Ratio (LWR) measures the proportion of tokens with seven or more characters, a length-independent proxy for technical or Latinate register.

5.2 Results

Document	Tokens	MSL	MWL	MATTR	PD	LWR
A06785	35,654	29.5	4.55	0.676	578	0.214
A06786	223,411	37.3	4.51	0.653	572	0.201
A06788	28,536	33.7	4.50	0.669	572	0.189
A06789	23,195	31.2	4.60	0.681	580	0.213
A06790	11,430	48.8	4.51	0.709	566	0.196
A06791	18,431	42.4	4.55	0.672	572	0.196
A07594 (Circle of Commerce)	33,601	20.9	4.44	0.664	569	0.197
A07886	14,106	26.8	4.47	0.708	575	0.184
A32827	49,656	47.2	4.44	0.693	562	0.194
A32828	5,758	48.0	4.66	0.690	581	0.238
A32829	4,232	41.5	4.50	0.690	572	0.198
A32830	2,939	40.3	4.45	0.714	562	0.198
A32833	49,725	46.6	4.44	0.693	562	0.194
A32836	4,772	48.2	4.38	0.684	555	0.194
A32837	3,076	31.4	4.23	0.678	542	0.175
A32838	2,970	39.6	4.57	0.687	590	0.203
A32839	11,861	28.2	4.46	0.695	569	0.191
A50763	5,204	48.6	4.40	0.680	557	0.195
A51598	27,722	54.1	4.50	0.717	563	0.199
A69858	5,759	22.9	4.48	0.694	577	0.191
A93819	2,586	41.7	4.49	0.665	567	0.200
B14801 (Free Trade)	19,775	24.9	4.56	0.668	573	0.210
Wealth	380,080	31.8	4.65	0.642	634	0.230

Table 2. Surface register complexity measures for all 23 documents.

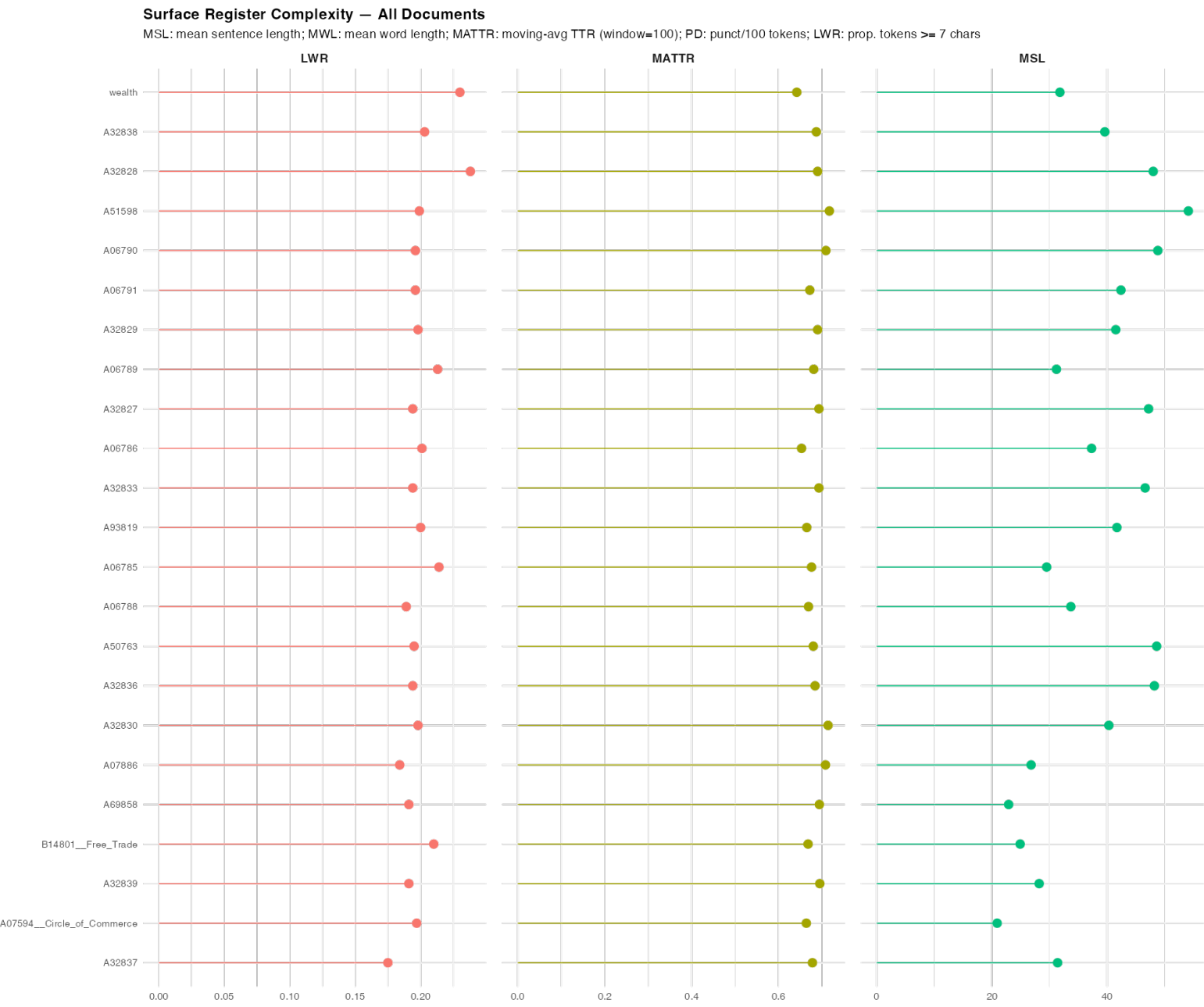


Figure 3. Surface register complexity across all 23 documents. Documents sorted by mean value within each panel.

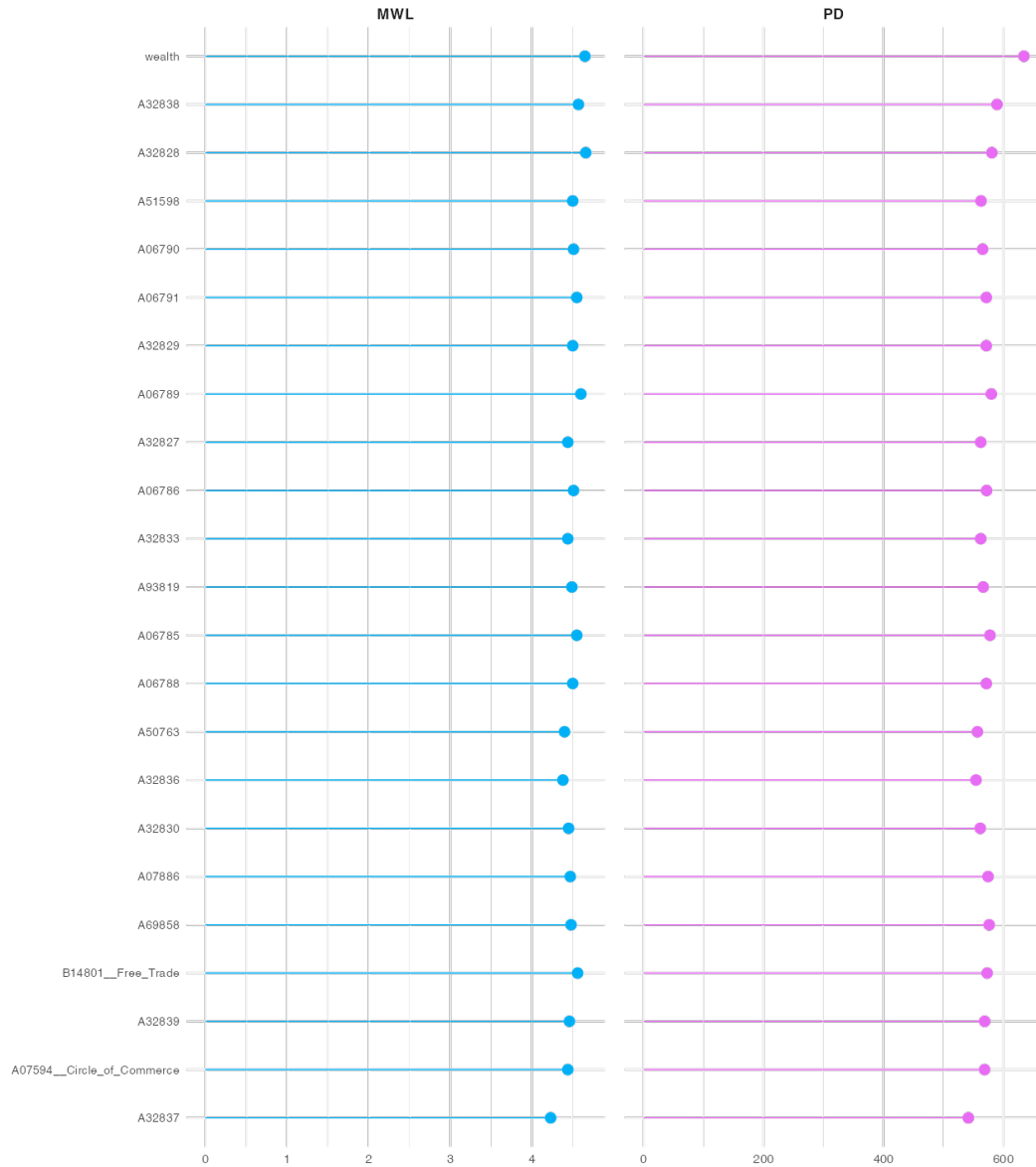


Figure 4. Surface register complexity across all 23 documents. Documents sorted by mean value within each panel.

5.3 Interpretation

1. MSL ranges from 20.9 (A07594, Misselden’s Circle of Commerce) to 54.1 (A51598), with most texts falling between 28 and 49. The three highest MSL scores belong to A51598 (54.1), A06790 (48.8), and A50763 (48.6). A51598’s extreme sentence length is consistent with its status as a merchant’s long-form discourse; A06790’s high MSL reflects the Canker’s allegorical prose style, which builds long accumulative sentences through parataxis rather than subordination. Misselden’s low MSL (20.9) is striking and interpretively significant: it suggests a more clipped, argumentative style with shorter, denser sentences more typical of polemical exchange than of the extended analytical mode of Malynes’s treatises.

2. MWL is remarkably consistent across the corpus, ranging only from 4.23 (A32837) to 4.66 (A32828), with most texts clustering tightly around 4.45–4.55. This narrow range reflects the shared early modern English register of most texts, where Latinate and Germanic vocabulary are in broadly similar proportions. The outliers are marginal: A32828's slightly elevated MWL likely reflects its administrative prose style, while A32837's lower score reflects simpler lexical choices in a shorter polemical pamphlet. The wealth document records the highest MWL at 4.65, consistent with Smith's more Latinate analytical prose.

3. MATTR ranges from 0.642 (wealth) to 0.717 (A51598), with A32830 also high at 0.714 and A07886 at 0.708. The lowest MATTR belongs to wealth (0.642) and A06786 (0.653). This is counterintuitive at first glance - one might expect the longest, most wide-ranging text to have the highest lexical diversity. However, low MATTR in wealth reflects sustained terminological repetition: Smith's analytical argument requires constant recurrence of key terms ('labor', 'capital', 'stock', 'price') across 380,000 tokens. A06786's low MATTR similarly reflects the accounting-table structure of *Lex Mercatoria*, which repeatedly cycles through the same set of currency and weight terms. High MATTR in A51598 and A32830 reflects their relative brevity and lexical variety within a focused argumentative frame.

4. PD shows one clear outlier: wealth records 634 punctuation marks per 100 tokens, substantially above all other texts (next highest: A32828 at 581, A06789 at 580). Smith's dense use of punctuation reflects his extended periodic sentences, which are heavily segmented with commas, semicolons, and parenthetical clauses. Among the early modern texts, PD is broadly consistent (542–581), suggesting that punctuation density was not a major differentiator of genre or register within this period.

5. LWR ranges from 0.175 (A32837) to 0.238 (A32828), with wealth at 0.230 and A06785 at 0.214. A32837's low LWR is consistent with its simple vocabulary (its top TF-IDF terms - 'usurer', 'usurers', 'richest' - are not especially long), while A32828's high LWR reflects administrative and Latinate vocabulary ('impediments', 'ministering', 'instrumental'). The wealth document's elevated LWR is consistent with its systematic use of abstract economic concepts.

6. Limitations and Methodological Reflections

Several limitations affect the analysis. First, the corpus is highly imbalanced: wealth (380,080 tokens) is over 130 times larger than A93819 (2,586 tokens). Document length affects TF-IDF in known ways - longer documents accumulate TF-IDF weight for a wider range of terms - and inflates absolute TF-IDF values even when the weighting formula partially accounts for document length. The near-duplicate status of A32827/A32833 and A32836/A50763 was confirmed by close inspection of both the TF-IDF terms and word counts; automated near-duplicate detection prior to analysis would have been preferable.

Second, variant spelling preservation, while analytically motivated, means that 'mony' and 'money' are counted as distinct types throughout. In a corpus where spelling is genuinely variable, this suppresses apparent similarity between texts that use the same concept with different orthography. A sensitivity analysis - rerunning the same procedures with and without spelling normalization - would establish whether and where this choice materially affects the results.

Third, the stopword list did not fully suppress high-frequency function words in all documents. The word ‘unto’ appears in A06790’s top TF-IDF terms (TF-IDF = 11.2), indicating it is unusually concentrated in that text relative to others, but it is still primarily a function word rather than a content term. This reflects a limitation of corpus-wide stopword lists when applied to documents with markedly different function-word distributions and suggests that per-document frequency thresholds might complement the stopword approach.

Fourth, the surface measures used in Approach 3, while parser-free, are not without limitations. Sentence segmentation based on sentence-ending punctuation is sensitive to OCR inconsistencies and early modern punctuation conventions, which do not always match modern expectations: a full stop in a seventeenth-century text may mark abbreviation rather than sentence end. MATTR variance increases for very short documents; the results for A32830 (2,939 tokens), A93819 (2,586 tokens), and A32837 (3,076 tokens) should be interpreted with greater caution than those for the larger texts.
